

translocation analysis - orientation

2025-09-30

load packages, read in processed dataset, normalize variables

```
library(tidyverse)
#library(NatParksPalettes)
library(scico)
library(viridis)
library(ggpubr)
library(readxl)
library(lme4)
library(car)

theme_set(theme_classic())

#edit to quantNorm function - remove NAs from estimation
quantNorm =function(x){
  x1<-x[!is.na(x)]
  x[!is.na(x)]<-qnorm(rank(x1,ties.method = "average")/(length(x1)+1))
  x
}

birds<-read_xlsx("C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/data_archiving/translocation_data.xlsx")
pembertonBirds<-read_xlsx("C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/data_archiving/translocation_data_pemberton.xlsx")

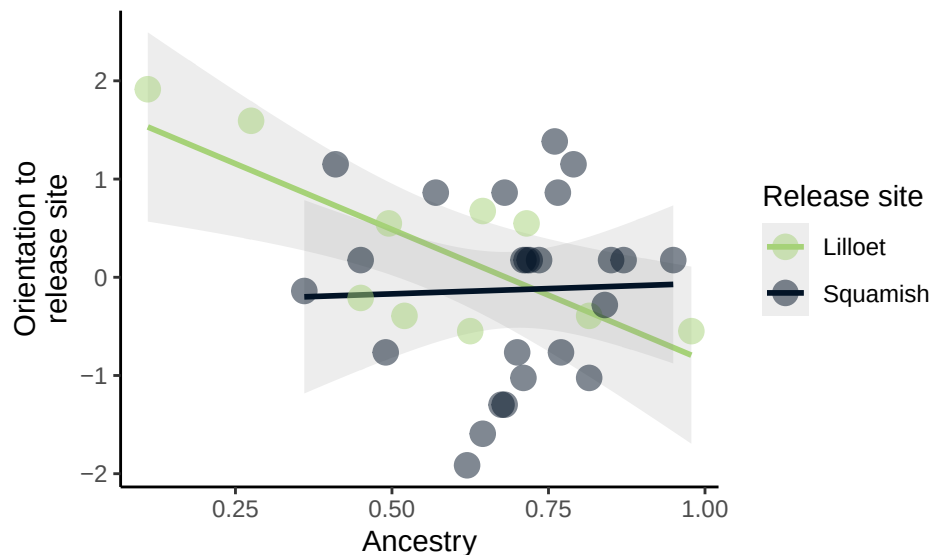
birds2<-rbind(pembertonBirds,birds)%>%
  mutate(fall_releaseSite_bearing_tier1=as.numeric(fall_releaseSite_bearing_tier1))%>%
  mutate(norm_releaseSite_bearing_1=quantNorm(fall_releaseSite_bearing_tier1))%>%
  mutate(ancestry=hiest_ancestry*(-1)+1)

birds<-birds%>%
  mutate(fall_releaseSite_bearing_tier1=as.numeric(fall_releaseSite_bearing_tier1))%>%
  mutate(norm_releaseSite_bearing_1=quantNorm(fall_releaseSite_bearing_tier1))%>%
  mutate(ancestry=hiest_ancestry*(-1)+1)
```

Orientations from Lilloet and Squamish

- expect effects of release site (environment), ancestry (genetics), and interaction between release site and environment (genetics x environment)
- orientation estimated in two ways: relative to the hybrid zone (i.e. Pemberton) and relative to each bird's release location

```
ggplot(birds,aes(x=ancestry,y=norm_releaseSite_bearing_1,
                group=release_site,
                colour=release_site))+
  geom_smooth(method="lm",alpha=0.18)+
  geom_point(size=4,shape=19,alpha=0.5)+
  scale_color_manual(values=scico(n=3,palette='navia',end=0.8)[c(3,1)],name="Release site")+
  xlab("Ancestry")+
  ylab("Orientation to\nrelease site")
```



```
lm2<-lm(norm_releaseSite_bearing_1~release_site*ancestry,data=birds,
        na.action=na.exclude)
anova(lm2)
```

```
## Analysis of Variance Table
##
## Response: norm_releaseSite_bearing_1
##
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
release_site	1	1.4247	1.42468	1.9045	0.17745
ancestry	1	1.9144	1.91445	2.5592	0.11979
release_site:ancestry	1	2.2417	2.24171	2.9967	0.09338
Residuals	31	23.1896	0.74805		

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(lm2)
```

```
##
## Call:
## lm(formula = norm_releaseSite_bearing_1 ~ release_site * ancestry,
##     data = birds, na.action = na.exclude)
##
## Residuals:
```

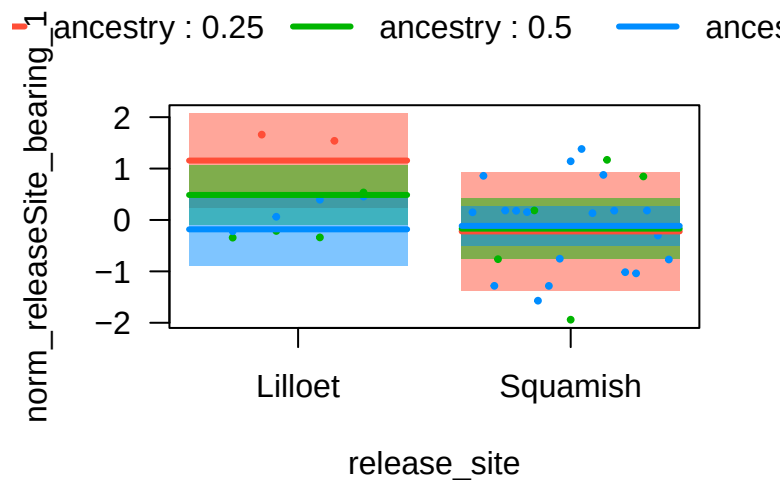
	Min	1Q	Median	3Q	Max

```
## -1.7713 -0.6772 0.2476 0.4441 1.4962
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.8250     0.6968   2.619  0.0135 *
## release_siteSquamish -2.1011     1.1082  -1.896  0.0673 .
## ancestry          -2.6765     1.1386  -2.351  0.0253 *
## release_siteSquamish:ancestry  2.8908     1.6699   1.731  0.0934 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8649 on 31 degrees of freedom
## (15 observations deleted due to missingness)
## Multiple R-squared:  0.194, Adjusted R-squared:  0.116
## F-statistic: 2.487 on 3 and 31 DF, p-value: 0.07892
```

```
emmeans::emmeans(lm2, ~release_site)
```

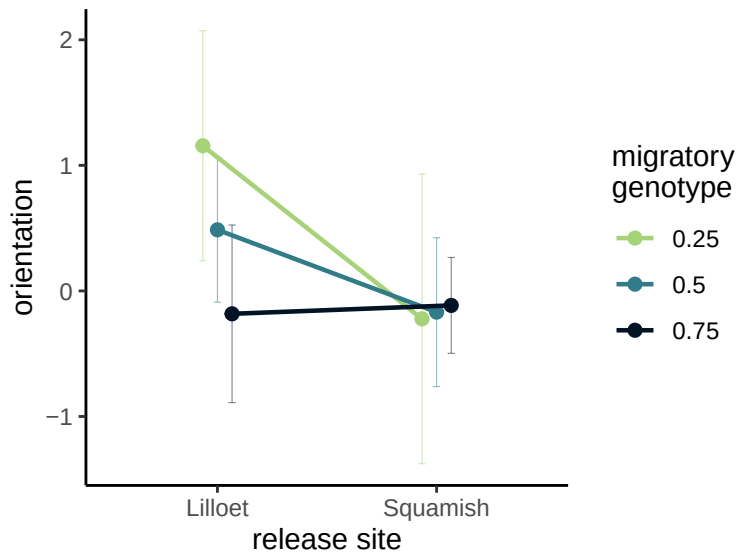
```
## release_site emmean SE df lower.CL upper.CL
## Lilloet      0.0732 0.293 31  -0.524  0.670
## Squamish     -0.1358 0.179 31  -0.500  0.229
##
## Confidence level used: 0.95
```

```
p=visreg::visreg(lm2,"release_site",by="ancestry",overlay=T,breaks=c(0.25,0.5,0.75))
```



```
gg1<-ggplot(p$fit, aes(release_site, visregFit,
                        colour=factor(ancestry),group=as.factor(ancestry))) +
  geom_errorbar(aes(ymin=visregLwr, ymax=visregUpr), alpha=0.5,
                linetype=1, size=0.2,width=0.1,position=position_dodge(width=0.2)) +
  geom_point(position=position_dodge(width=0.2),size=2) +
  geom_line(position=position_dodge(width=0.2),size=0.8)+
  scale_color_manual(values=scico(n=3,palette='navia',end=0.8,direction=-1),name="migratory\ngenotype")
```

```
labs(fill="Weight")+
ylab("orientation")+xlab("release site")
gg1
```



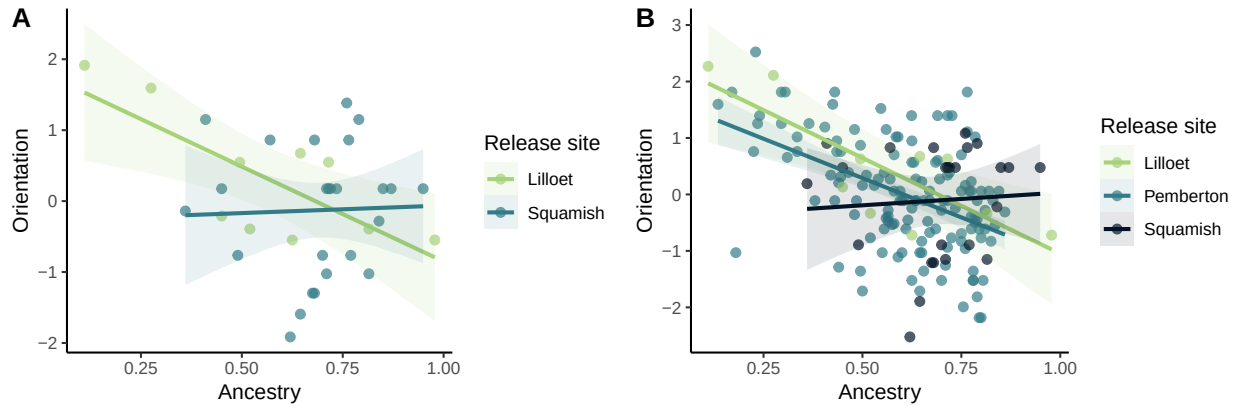
Comparison to Pemberton birds

- selected birds released in Pemberton in 2020 & 2023
- bearing shows a negative relationship to ancestry - inland birds fly east, coastal birds fly west

```
p1<-ggplot(birds,aes(x=ancestry,y=norm_releaseSite_bearing_1,
group=release_site,
colour=release_site,fill=release_site))+
geom_point(size=2,shape=19,alpha=0.7)+
geom_smooth(method="lm",alpha=0.1)+
scale_color_manual(values=scico(n=3,palette='navia',end=0.8,direction=-1),name="Release site")+
scale_fill_manual(values=scico(n=3,palette='navia',end=0.8,direction=-1),name="Release site")+
xlab("Ancestry")+
ylab("Orientation")

p2<-ggplot(birds2,aes(x=ancestry,y=norm_releaseSite_bearing_1,
group=release_site,
colour=release_site,fill=release_site))+
geom_point(size=2,shape=19,alpha=0.7)+
geom_smooth(method="lm",alpha=0.1)+
scale_color_manual(values=scico(n=3,palette='navia',end=0.8,direction=-1),name="Release site")+
scale_fill_manual(values=scico(n=3,palette='navia',end=0.8,direction=-1),name="Release site")+
xlab("Ancestry")+
ylab("Orientation")

ggarrange(p1,p2,labels=LETTERS)
```



```
#ggsave("C:/Users/Steph/TAMU_OneDrive//Thrushes/translocation/FigS1.png",
#       width=9,height=3)
```

- include year as a random effect, to control for similarities among 2020 vs 2023 birds
- here, release site is unimportant, but ancestry is significant and there is an interaction between release site and ancestry
- again, more positive slopes between release site and ancestry in Lilloet and Squamish than Pemberton

```
birds2<-birds2%>%mutate(f_release_year=as.factor(release_year))%>%
  mutate(f_release_site=factor(release_site,levels=c("Squamish","Pemberton","Lilloet")))

lmer2<-lmer(norm_releaseSite_bearing_1~f_release_site*ancestry+(1|f_release_year),
            data=birds2,na.action=na.exclude)
Anova(lmer2)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: norm_releaseSite_bearing_1
##              Chisq Df Pr(>Chisq)
## f_release_site    2.3686  2    0.30595
## ancestry          43.8792  1  3.493e-11 ***
## f_release_site:ancestry  7.2696  2    0.02639 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coef(lmer2)
```

```
## $f_release_year
##      (Intercept) f_release_sitePemberton f_release_siteLilloet  ancestry
## 2020  -0.6622228          2.285777          3.002956  0.4519234
## 2023  -0.4185420          2.285777          3.002956  0.4519234
##      f_release_sitePemberton:ancestry f_release_siteLilloet:ancestry
## 2020          -3.340889          -3.840826
## 2023          -3.340889          -3.840826
##
## attr(,"class")
## [1] "coef.mer"
```

```
summary(lmer2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: norm_releaseSite_bearing_1 ~ f_release_site * ancestry + (1 |
##   f_release_year)
##   Data: birds2
##
## REML criterion at convergence: 429.3
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -2.7833 -0.6633  0.1184  0.6688  2.5165
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## f_release_year (Intercept) 0.03824   0.1956
## Residual                0.73419   0.8568
## Number of obs: 171, groups: f_release_year, 2
##
## Fixed effects:
##
##              Estimate Std. Error t value
## (Intercept)      -0.5404     0.8674  -0.623
## f_release_sitePemberton    2.2858     0.9037   2.529
## f_release_siteLilloet     3.0030     1.1057   2.716
## ancestry           0.4519     1.2101   0.373
## f_release_sitePemberton:ancestry -3.3409     1.2896  -2.591
## f_release_siteLilloet:ancestry -3.8408     1.6543  -2.322
##
## Correlation of Fixed Effects:
##              (Intr) f_rl_P f_rl_L ancstr f_r_P:
## f_rls_stPmb -0.938
## f_rls_stLll -0.769  0.743
## ancestry   -0.964  0.926  0.756
## f_rls_stPm:  0.908 -0.975 -0.715 -0.938
## f_rls_stLl:  0.705 -0.677 -0.945 -0.731  0.686
```

```
data.frame(Anova(lmer2))%>%
  mutate()
```

```
##              Chisq Df    Pr..Chisq.
## f_release_site    2.368640  2 3.059542e-01
## ancestry         43.879246  1 3.492728e-11
## f_release_site:ancestry 7.269611  2 2.638907e-02
```

```
lm2b<-lm(norm_releaseSite_bearing_1~f_release_year+f_release_site*ancestry,
          data=birds2,na.action=na.exclude)
Anova(lm2b)
```

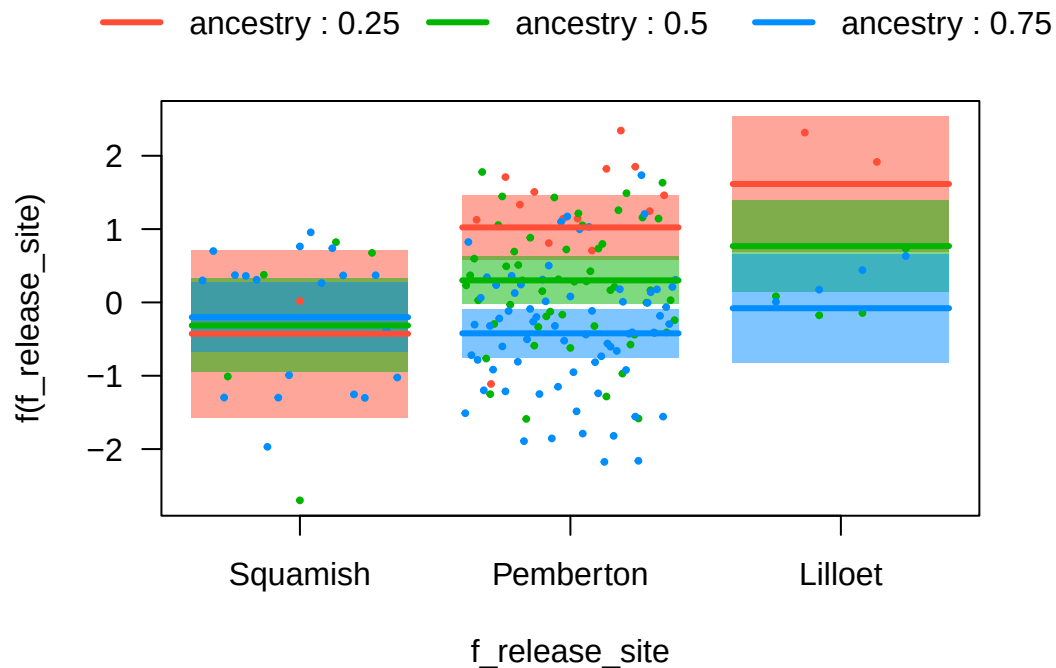
```
## Anova Table (Type II tests)
##
## Response: norm_releaseSite_bearing_1
##              Sum Sq  Df F value    Pr(>F)
```

```
## f_release_year      3.283    1  4.4716    0.03597 *
## f_release_site      1.987    2  1.3532    0.26128
## ancestry            32.652    1 44.4730  3.761e-10 ***
## f_release_site:ancestry  5.401    2  3.6779    0.02738 *
## Residuals          120.407  164
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lm2c<-lm(norm_releaseSite_bearing_1~f_release_year*ancestry,
          data=birds2%>%filter(release_site=="Pemberton"),
          na.action=na.exclude)
Anova(lm2c)
```

```
## Anova Table (Type II tests)
##
## Response: norm_releaseSite_bearing_1
##              Sum Sq Df F value    Pr(>F)
## f_release_year      3.283    1  4.7670    0.03078 *
## ancestry            31.323    1 45.4822  4.377e-10 ***
## f_release_year:ancestry  1.578    1  2.2906    0.13255
## Residuals          90.908  132
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

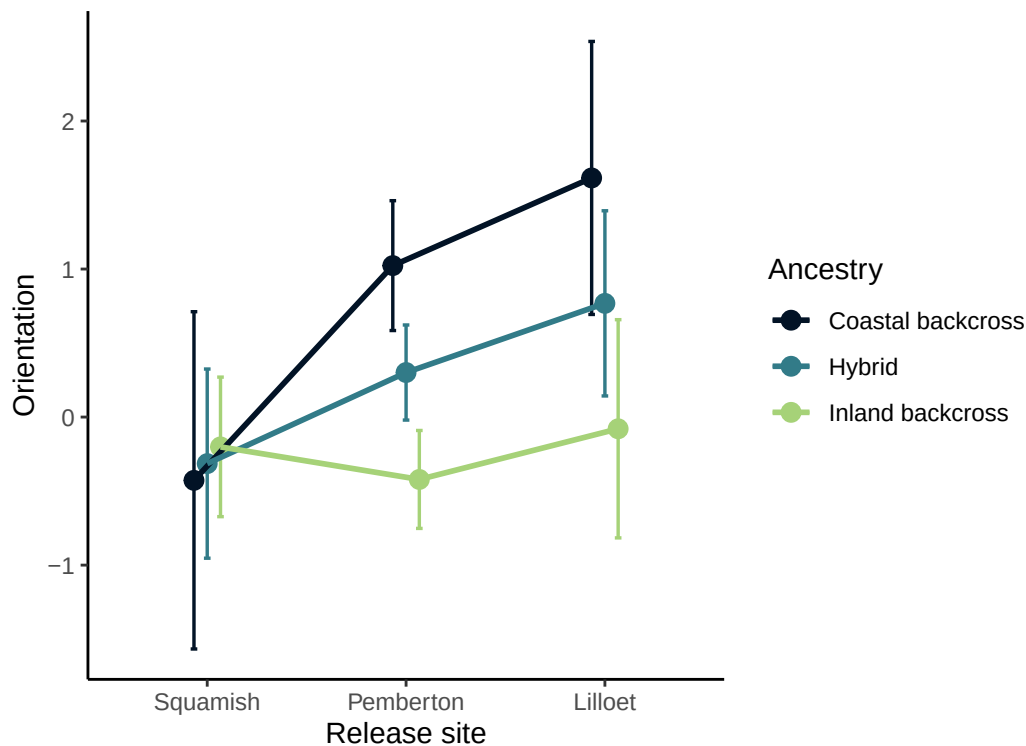
```
p=visreg::visreg(lmer2,"f_release_site",by="ancestry",overlay=T,breaks=c(0.25,0.5,0.75))
```



```

gg1<-ggplot(p$fit, aes(f_release_site,
                      visregFit,
                      colour=factor(ancestry),group=as.factor(ancestry))) +
  geom_errorbar(aes(ymin=visregLwr, ymax=visregUpr),
               linetype=1, size=0.6,width=0.1,position=position_dodge(width=0.2)) +
  geom_point(position=position_dodge(width=0.2),size=3) +
  geom_line(position=position_dodge(width=0.2),size=1)+
  scale_color_manual(values=scico(n=3,palette='navia',end=0.8),name="Ancestry",
                    labels=c("Coastal backcross","Hybrid","Inland backcross"))+
  labs(fill="Weight")+
  ylab("Orientation")+xlab("Release site")
gg1

```



```

#ggsave(plot=gg1,
#       file="C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/Fig3B.svg",
#       width=5.5,height=4)

```